Commodity Hedging & Structuring Workbench

A Learning Reference and Validation Report

learner workbench (extension of the Citi MQA Forage simulation)

September 2026

Executive summary

Validation verdict: six of seven outcome checks PASS. The one FLAG (historical 95% VaR coverage of 38 breaches versus 25.2 expected under the static historical estimator) is retained and discussed; the filtered (EWMA-scaled) estimator remediated it, covering at 5.5% of days versus the 5% expectation (ratio 1.10, inside the pre-declared band).

This report closes the validation phase of a practitioner-style commodity hedging workbench built as an extension of the Citi MQA Forage simulation. The workbench prices and hedges a roaster's coffee purchase program on a frozen ICE Coffee C (KC=F) curve: data gates, implied convenience-yield term structure, a Schwartz–Smith two-factor fit, GARCH(1,1) working volatility, a roaster futures-and-collar hedge program, a capital-protected participation note, and a junior xVA layer (EE/EPE/PFE, unilateral CVA, netting and collateral).

Phase 6 adds two things. First, an SR 26-2-aligned validation harness: a model inventory, a fail-closed no-look-ahead gate on calibration windows, and a rolling-window outcomes program across every model. Second, the decision analysis: the roaster's outcome unhedged versus futures-only versus a zero-cost collar, with margin-funding peaks per strategy.

Headline decision numbers (1.2M lb annual purchases, 2.67 average contracts, front price 324.25 ¢/lb): the full futures hedge locks the annual purchase cost at $3,891,000 with a 95th-percentile margin peak of $103,797; the zero-cost collar keeps the cost locked between the strikes with a materially lower margin peak ($44,250 at the 95th percentile) at the price of re-exposure beyond the strikes; the unhedged roaster faces a 95th-percentile cost of $5,288,885 and a simulated worst case of $10.3M.

All regulation content (SR 26-2, IFRS 9, EMIR 3, MiFID II/PRIIPs) is *mapped as methodology*, never asserted as compliance. All numbers in this report are regenerated in one run by `report/make_numbers.py` from the frozen data (SHA-256 manifests); the hand-calculated checks of Section 11 are reproducible without reading code.

Part I
Concepts and methodology: how the workbench works

Part I explains every concept the workbench uses, in the order a practitioner meets them: the business problem, the data, the curve, the price model, the hedge, the note, the counterparty risk, and the regulation that shapes each. Every concept follows the same pattern: what it is, why a desk needs it, the formula, a worked number from the frozen data, and the interpretation. Part II then reports what was validated and what the numbers decided.

1 The business problem

A coffee roaster buys green coffee continuously. Our reference program is 100,000 lb per month for twelve months (1.2M lb total), priced off the ICE Coffee C futures contract (37,500 lb per contract). Coffee can move tens of percent in a year; the roaster's retail margin cannot. Left unhedged, the annual purchase cost of this program has a simulated 95th-percentile of \$5.29M and a worst case of \$10.3M against a forward-implied \$3.89M — that gap is the risk the workbench prices, hedges, and explains.

Forward and futures contracts

A *forward* is a bilateral agreement to exchange at a fixed price on a fixed date; a *futures* contract is the exchange-traded equivalent, standardized in size (\$37,500 per 1 cent move per coffee contract), expiration months, and quality. Two differences matter here. First, futures are cleared through a central counterparty, so the roaster's credit exposure to any single dealer is replaced by margin obligations to the clearing house. Second, futures are marked to market daily: gains and losses settle in cash every day (variation margin). The daily settlement is why margin funding — not just the final price — becomes a risk to manage.

Basis

The roaster's physical purchases settle at an origin differential over the exchange C price, not at the exchange price itself. *Basis* is the gap between the local physical price and the futures reference. A futures hedge removes the exchange-price risk; the basis remains in the roaster's P&L. The workbench evaluates every hedge on the exchange series and says so: basis is not designated in the hedge relationship and not modelled (Section 14).

Backwardation and contango

A futures *curve* is a set of prices by expiry month. *Backwardation* means deferred months trade below the front; *contango* is the reverse. On the frozen curve (as-of 2026-09-04) every consecutive spread is backwardated: the front (Sep 2026) is 324.25 ¢/lb and each later month sits lower. A backwardated curve is the commodity analogue of a stock paying a large dividend: something valuable accrues to holders of the physical good.

Convenience yield

That "something" is the *convenience yield*: the benefit of holding physical inventory (ability to ship now, keep roasting lines fed, exploit local scarcity) beyond its financing and storage cost. When near coffee is scarce, users bid up the front of the curve and the implied convenience yield rises. Chapter 3 extracts it from the curve; the frozen curve implies 40.8%/yr at the front, decaying to 6.3%/yr on the back — the market is paying heavily for immediate availability.

Why hedge at all: the unhedged cost distribution above is one tail away from a bad year. The rest of Part I builds, piece by piece, the models that turn market data into that distribution and into hedge structures that change it.

2 Data engineering as a model-risk control

Every number in this workbench traces to a small set of frozen files. That is deliberate: in model-risk terms, uncontrolled input data is itself a model risk, and the cheapest control is a reproducible, checksummed snapshot.

Data freeze with SHA-256 manifests

The downloader writes each series to `data/frozen/`, computes a SHA-256 hash per file, and records it in a JSON manifest. Every later use of the data re-verifies the hash before reading. A changed file is not silently accepted; the pipeline stops. This is what makes "every number regenerated in one run" a verifiable claim rather than a hope: the manifests are committed, so any reader can confirm the exact bytes behind the results.

No-look-ahead gate

A *look-ahead error* occurs when a calibration window accidentally sees data after its valuation as-of date; it silently inflates every backtest. The workbench makes the failure impossible to miss: each calibration call passes through a gate that truncates the series to the as-of window and raises a `LookaheadError` if any observation past the as-of date is requested. The gate is itself tested by feeding it a poisoned window and asserting it refuses.

Fail-closed design with a pre-declared fallback

The universe logic is *fail-closed*: coffee gate failures switch to the pre-declared gold fallback; a gold failure stops everything and nothing downstream is produced. The opposite (fail-open) design, returning partial or stale data on failure, is how mismatched numbers enter production systems unnoticed. Declaring the fallback *before* it is needed prevents the subtle failure where the fallback is chosen because it gives the preferred answer.

The sources are stated plainly: Yahoo Finance daily bars (an unofficial source, accepted and documented) for the ICE contract chain, and the SOFR rate from FRED [13]. The manifests, gate reports, and executed notebooks are committed; the price bars are not (exchange-data licensing, Section 9) and regenerate locally with one command.

3 The forward curve and cost of carry

The curve is where commodity desks start. This chapter derives the one identity that turns two futures prices into an implied yield, applies it to the frozen coffee curve, and reads what it says about the physical market.

The cost-of-carry identity, derived

Suppose you buy coffee now at spot S, pay financing at rate r and storage at rate d, and hold one unit for a period τ; a futures sale at F locks the outcome. No-arbitrage pins the two choices together: buying and carrying must cost the same as buying forward, so $F = S\,e^{(r+d-y)\tau}$, where y is the annual convenience yield introduced in Chapter 1. The convenience yield enters with a minus sign: holding the physical good earns the convenience benefit, so the forward only has to compensate financing plus storage *minus* that benefit. Now take the ratio of two adjacent futures prices F_1 (near) and F_2 (next expiry, τ apart). Spot, S, cancels, as does everything not known between the two dates:

$$\frac{\ln F_2 - \ln F_1}{\tau} \;=\; r + d - y \qquad \Longrightarrow \qquad y \;=\; r + d - \frac{\ln(F_2/F_1)}{\tau}.$$

Two adjacent futures prices and a financing rate are enough to extract the market's own implied convenience yield.

Worked on the frozen curve

As of 2026-09-04 the coffee chain carries 8 contract months plus the continuous front (KC=F). With SOFR frozen at 3.66% and storage assumed zero (see the caveat below), applying the identity spread by spread gives an implied convenience yield of 40.8%/yr at the front, decaying to 6.3%/yr on the back, with the curve backwardated at every consecutive spread. The package computes this as `carry.implied_yield`; the curve-stability outcomes check re-runs it at six historical as-of dates (front yield 14.8–40.8%/yr; backwardated at 4 of 6, with two historical contango episodes preserved rather than tuned away).

Interpretation. A 40%/yr implied convenience yield says the market prices immediate availability as scarce: hedgers who need physical coffee now, or who fear not having it, bid the front up relative to deferred months. The decay from front to back is normal — scarcity is a short-dated phenomenon — and the backwardated shape is the term structure signature of that scarcity. For the roaster program this matters directly: rolling a long position through a backwardated curve (*rolling up* the curve) tends to buy high and sell low each cycle, a recurring carry cost the hedge program must budget for.

Storage is assumed, not observed

The identity reports y *gross of storage*: separating convenience yield from storage outlay d needs a storage-cost assumption the data does not carry, so the workbench fixes $d = 0$ and states it. Reported "convenience yield" is therefore an upper bound on the true yield by the size of the true storage rate. The honest fix is not a secret calibration but a stated assumption.

An unofficial source can distort a yield

Yahoo prices are unofficial; a stale or inconsistent quote between contract months would show up as an implausible implied yield. This is why the curve-stability check exists: it re-derives the yield at six as-of dates and asserts the front yield stays inside a pre-declared plausibility band. A data problem announces itself as a curve problem.

4 Price dynamics: level and volatility

Three later chapters need a price process: the exposure engine simulates future MtM, the Monte Carlo prices option payoffs, and the hedge analysis stress-tests funding peaks. This chapter covers the two models the workbench fits — a two-factor level model and a volatility model — and the discipline of choosing between them with pre-declared criteria.

4.1 Level: the Schwartz–Smith two-factor model

Two factors, two horizons

Commodity prices mix two behaviors: short-term mean reversion (weather, supply shocks, inventory swings wash out) and long-term drift in the equilibrium price level (cost inflation, demand cycles). Schwartz and Smith [10] model the log price as the sum of both: $S_t = \exp(X_t + \chi_t)$, where X_t is the long-term factor (a geometric walk with equilibrium level ξ and reversion speed κ) and χ_t is the short-term factor (an Ornstein–Uhlenbeck deviation that decays toward zero at speed κ). The mean-reversion speed has a natural unit: the *half-life* $\ln 2/\kappa$, the time for half of a deviation to disappear.

Fit and interpretation on the frozen curve

The reduced-form fit on the frozen curve gives mean-reversion speed $\kappa = 5.97$/yr, half-life $\ln 2/5.97 \approx 0.116$ yr (≈ 1.4 months), and log equilibrium level $\xi = 5.673$, i.e. $e^{\xi} \approx 291$ ¢/lb. The fit RMSE sits far inside the pre-declared tolerance. Interpretation: the market believes coffee reverts to roughly 291 ¢/lb and that price surprises are absorbed in weeks, not months — while the current front (324.25) stands above the equilibrium, consistent with the scarcity reading of the 40% front convenience yield.

Reduced form, single snapshot

The fit uses one curve snapshot, which identifies the level parameters (κ, ξ) but not the volatility split between the two factors; a full stochastic fit needs Kalman maximum-likelihood estimation on the whole panel of maturities. The workbench flags this rather than hiding it: the fit is used for level interpretation, and every premium in Part II carries a GARCH volatility estimate on top of it.

4.2 Volatility: EWMA and GARCH(1,1)

EWMA — the RiskMetrics estimator

The exponentially weighted moving average [8] updates yesterday's variance with today's squared return, $\sigma_t^2 = \lambda\,\sigma_{t-1}^2 + (1-\lambda)\,r_{t-1}^2$, with decay $\lambda = 0.94$ on daily data. It is simple, always defined, and reacts fast to volatility clustering — but it has no long-run level: it is a GARCH(1,1) with persistence forced to 1, so it can drift far from anything sustainable. On the frozen returns the EWMA reads 47.2%/yr.

GARCH(1,1) — the workhorse

The GARCH(1,1) of Bollerslev [2] adds mean reversion: $\sigma_t^2 = \omega + \alpha\, r_{t-1}^2 + \beta\, \sigma_{t-1}^2$. The long-run variance is $\omega/(1-\alpha-\beta)$, and $\alpha+\beta$ is the *persistence* — how slowly variance decays back to it. On the frozen data: persistence 0.802, long-run level 38.5%/yr, last conditional estimate 39.2%/yr. The current level sits almost exactly at its own long-run mean — the market is in a normal-volatility regime.

Choosing with a pre-declared criterion

Model choice is an outcome, not a preference: the workbench declares before looking that GARCH is adopted only if its rolling one-step forecast RMSE beats the EWMA reference with ratio ≤ 1.1. On four rolling windows the RMSEs are 4.01 vs 6.80%/yr (ratio 0.59) — GARCH wins clearly and is adopted as the working volatility. The EWMA stays as the benchmark model in the inventory, which is exactly how an SR 26-2-style outcomes analysis should treat a model selection (Section 8).

Historical volatility is not implied volatility

Every premium and VaR below inherits this one number from frozen history; parameter risk flows through. No free implied-volatility surface exists for coffee options, so there is no market check on the level — stated as a limitation in Section 14, not fixed by hoping.

5 Hedging the roaster program

With a curve, a carry reading, and a volatility model, the roaster can actually hedge. This chapter builds the two candidate structures — the plain futures lock and the zero-cost collar — and states exactly what each one does to the cost distribution.

5.1 The futures hedge and its plumbing

The program is 1.2M lb over 12 months, i.e. 32 contract-months (1.2M/37,500), an average of 2.67 contracts outstanding. The hedge selects each month's contract with the *in-or-after* rule (the expiry month at or after the physical delivery date) and rolls each contract 10 trading days before expiry, so no position is forced into delivery.

Margin: the liquidity side of hedging

Posting the hedge requires *initial margin* (performance deposit, $8k per contract here — a stated learner assumption, not an exchange feed) and daily *variation margin*: every day the futures lose, cash leaves. A perfectly correct hedge can still bankrupt its owner in the middle if the price moves far enough before it pays off. This is why the workbench reports *margin-funding peaks* (the running cumulative cash drain) alongside every cost number: the futures hedge locks the price at $3,891,000 but carries a 95th-percentile margin peak of $103,797 that must be funded throughout.

5.2 The zero-cost collar

Construction

Buy the futures hedge, buy a put struck at $K_p = 280$, and sell a call struck at $K_c = 382.06$ with the premium equal to the put's (both 7.69 ¢/lb; the strike K_c is solved so the premiums net to zero — the residual gap is 1.3×10^{-14} by construction). The put pays $(K_p - f_T)^+$ dollars per lb at expiry; the short call costs $(f_T - K_c)^+$. Per pound, the terminal cost is therefore

$$\text{cost}(f_T) = \begin{cases} f_0 - K_p + f_T & f_T < K_p \quad \text{(put monetized, cost floats but below the lock)} \\ f_0 & K_p \le f_T \le K_c \quad \text{(cost locked at the forward)} \\ f_0 - K_c + f_T & f_T > K_c \quad \text{(short call loses, cost floats)} \end{cases}$$

and the hedge P&L is capped between the strikes, which caps the *margin funding* at the put protection: $(324.25 - 280) \times \frac{8}{3} \times 375 = \$44{,}250$ at the 95th percentile, versus \$103,797 for the futures lock.

The strikes bound the hedge P&L, not the buyer's cost

"Locked between the strikes" is easy to misread. The *hedge position* gains are capped; the *buyer's total cost* still floats 1:1 with the market outside them: below 280 the monetized put drags the cost back down (a crash keeps helping), above 382.06 the short call bleeds (a spike keeps hurting). The collar is not a budget guarantee; it is a trade that trades tail cost for liquidity — p95 cost \$4.60M versus the futures' locked \$3.89M, with margin peaks roughly halved.

Choosing between them. For a roaster whose binding constraint is cash-flow certainty, pick the futures lock: same locked cost, full margin exposure. For a margin-constrained buyer, the collar halves the funding peak at the price of re-exposure beyond the strikes. Both structures are priced in the decision section (Part II) on the same scenario engine, so the comparison is like-for-like. Hedge accounting for either is treated in Section 8 (IFRS 9); the clearing-threshold classification in the EMIR 3 part of the same section.

6 The structured participation note

Structured notes are where curve, volatility, and discounting meet in one instrument. The workbench builds the desk-side version of the product the Citi Forage simulation gestures at: a note issued to an investor that returns principal plus a share of the coffee rally.

Capital-protected participation payoff

The note pays, per \$1 of notional at maturity T:

$$1 + p \max\left(\frac{F_T}{F_0} - 1,\ 0\right), \qquad p = 0.6,$$

a 9-month ATM participation on KC=F. The investor keeps the principal in every scenario and keeps 60% of any rally. This decomposes by design into two ordinary instruments: a zero-coupon bond (the principal) plus p calls struck at the forward F_0.

Black-76 pricing

Options on futures are priced by Black's model [1]: the premium is the discounted expectation under the risk-neutral measure in which the futures is a martingale,

$$c = e^{-rT}\big(F\,N(d_1) - K\,N(d_2)\big), \qquad d_1 = \frac{\ln(F/K) + \sigma^2 T/2}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T},$$

with the GARCH working volatility σ from Chapter 4 and r the frozen SOFR. The premiums of both collar legs in Chapter 5 come from the same formula, which is why a zero-premium collar is solvable at all.

Worked decomposition

At $F_0 = 324.25$, 9-month tenor, SOFR 3.66%: the discount bond is $e^{-rT} = 0.97292$; the ATM Black-76 call is 0.13128, so the participation piece is $0.6 \times 0.13128 = 0.07877$. Fair value: $0.97292 + 0.07877 = 1.05169$ per unit notional. An investor paying 105.17% of notional is financing 2.92 points of protection with 60% of the upside.

Monte Carlo as an independent check

The closed form assumes its own inputs; an implementation can still be wrong. The workbench re-prices the note with an antithetic Monte Carlo engine (100k paths, variance reduced by pairing each path's price with its mirror image): 1.051913 ± 0.000452, against the closed form at 1.051689 — a difference of 8.2×10^{-5}, under 2 standard errors. Two engines, two code paths, one answer: that two-way match is the trust mechanism, and it is asserted as an outcomes check in Part II.

One model, one premium

Both the collar legs and the note inherit the GARCH point estimate; there is no implied surface to cross-check against (no free listed coffee options chain). The issuer-side greeks (delta, vega) are validated by bump-and-reprice rather than market calibration. Parameter risk is declared, not eliminated.

Regulatory placement: a note like this sold to retail in the EU would require a PRIIPs key information document (Section 8); here the note is priced, never offered.

7 Counterparty risk and xVA

Any issuer or dealer holding the other side of these positions carries credit exposure: if the counterparty defaults while the trade is in our favor, we recover less than full value. The junior xVA layer quantifies that exposure and its price.

Exposure simulation: EE, EPE, PFE

The engine simulates the futures price under the risk-neutral measure as a lognormal martingale (the same assumption Black-76 prices), marks the book's MtM on every path, and collects, per future date t, the distribution of $\max(\mathrm{MtM}, 0)$ — the exposure only matters when we are owed money. Three summaries:

$$\mathrm{EE}(t) = \mathbb{E}\big[\max(\mathrm{MtM}(t), 0)\big], \quad \mathrm{EPE} = \tfrac{1}{T}\int_0^T \mathrm{EE}(t)\,dt, \quad \mathrm{PFE}_q(t) = q\text{-quantile of } \max(\mathrm{MtM}(t), 0).$$

EPE is the average exposure (what CVA prices); PFE is the tail exposure (what collateral limits are set against). By construction $\mathrm{EE}(t) \leq \mathrm{PFE}(t)$, and the suite asserts it. On the 2.67-contract book: EPE $33,755; 1-year PFE $250,975.

Unilateral CVA, bucketed

Credit valuation adjustment is the discounted expected loss from counterparty default, integrating exposure over time:

$$\mathrm{CVA} \;=\; \mathrm{LGD} \int_0^T e^{-rt}\, \mathrm{EE}(t)\; h\, e^{-ht}\, dt,$$

with default hazard h (flat 3% here) and loss given default LGD = 60%. The integral is evaluated on discrete time buckets (the $h\,e^{-ht}$ term is the per-bucket default density). On this book: $583.79. The engine is benchmarked by re-quoting a matched EE grid through QuantLib [12] — agreement to 1.9×10^{-16} relative on that grid (Section 12); the notebook's matched pair on this same engine (notebook 04) agrees to 4.8×10^{-16}.

Netting

Under a netting agreement, exposure to a set of positions is the net MtM of the set, not the sum of gross exposures: offsetting positions cancel before default law looks at them. Consequence: netted CVA $\leq$ sum of standalone CVAs, asserted in the suite. For the roaster program the netting set is the futures positions across contract months. At desk scale this is why ISDA netting collapses multibillion gross exposures to a fraction.

Collateral

Collateral — a threshold (unsecured exposure the counterparty may build before posting) plus independent amounts (posting required in advance) — attenuates the EE profile monotonically: every threshold unit cuts the tail exactly once, everywhere. Collateral limits are set against PFE, not EE: a limit sized on the average exposure is breached by half the paths. Thresholds also transform CVA, and the transform is directional (less unsecured exposure, less CVA) but not uniform.

Wrong-way risk, read rather than modelled

WWR is correlation between counterparty default and exposure. Here it is benign by inspection: the roaster buys physical coffee and hedges long, so its exposure to us grows when coffee rises — while the roaster's own business (it is a buyer of coffee) is stressed when coffee rises. That is *right*-way risk for us as the issuer of the note's short call. The dangerous mirror is selling the participation note's upside to a producer long coffee: exposure and counterparty distress would then rise together. Modelled WWR needs a correlated default–price process; kept qualitative here per scope, stated as such.

Junior xVA only

No DVA, FVA, or MVA; flat hazard and flat SOFR discounting. These omissions matter at dealer scale (funding costs of the hedge itself, own-default leg of the CVA). The layer is sized for a junior desk review, and says so.

8 Regulatory context

Four regimes shape what this workbench does. Each is treated the same way: the problem the regime exists to solve, what it requires, and how the workbench maps to it — mapped as methodology, never asserted as compliance, because a learner artifact has no regulatory status to assert.

8.1 SR 26-2: model risk management

The problem and the guidance

Models simplify reality; the gap between model and reality is model risk — the potential for loss or bad decisions when outputs are wrong or misused. The US banking agencies first codified expectations in SR 11-7 (2011) and issued the revised guidance SR 26-2 on 2026-04-17, superseding SR 11-7 and the BSA/AML addendum SR 21-8 [3]. The revised guidance is explicitly risk-based and tailored: it is expected to be most relevant to banking organizations above $30B in assets, and it sets no enforceable standards.

What SR 26-2 actually requires

A *model* is a quantitative method applying statistical, economic, or financial theory to process input data into estimates — simple arithmetic and rule-based software are excluded. Requirements, by lifecycle stage:

- *Development*: clear statement of purpose, testing out-of-sample and out-of-time, sensitivity to assumptions.
- *Validation*: (i) *conceptual soundness* — assess the design, assumptions, and judgments; (ii) *outcomes analysis* — compare model outputs to real-world outcomes (backtests, benchmarks); (iii) *ongoing monitoring* — detect performance deterioration.
- *Governance*: roles and independence (*effective challenge* by qualified, objective reviewers), a comprehensive model inventory, documentation adequate to understand each model's risk.
- *Aggregate view*: risk assessed across models together — common assumptions and shared data create correlated failures.
- *Vendor products*: third-party models are validated like in-house ones.

Mapping. The workbench reproduces the *mechanics* on a small scale: a 13-model inventory (Part II, Table 12.1), a rolling outcomes program across every model, benchmarking (GARCH vs EWMA, MC vs closed form, CVA vs QuantLib), fail-closed data gates, and hand-calculated checks in test docstrings as documentation. The posture is methodological fidelity: practice the discipline, claim no status.

8.2 IFRS 9: hedge accounting

The problem

Without hedge accounting, a hedging futures position marks to market in P&L *today* while the hedged purchase hits earnings *later*; earnings volatility appears purely from timing. Hedge accounting aligns the two by parking the effective portion of the hedge's gain or loss in other comprehensive income until the hedged transaction affects earnings.

Cash-flow hedge designation and effectiveness

IFRS 9 [4] permits designation of a forecast transaction as a hedged item when it is *highly probable* and presents a particular risk. For the roaster program: highly probable forecast green-coffee purchases (variable price), with coffee C price risk designated — the basis between origin differentials and the exchange C price is *not* designated and stays in P&L. Effectiveness is principles-based: there must be (i) an economic relationship between hedged item and instrument, (ii) credit risk not dominating value changes, and (iii) a hedge ratio consistent with the risk-management objective — the retired IAS 39 80–125% dollar-offset band is gone. The effective portion accumulates in AOCI and is reclassified to P&L in the period the forecast purchases affect earnings. Variation margin posted to the clearing house is a receivable/liability, not a hedge-accounting item.

Mapping. The program is assessed with `effectiveness.assess` on the hedge P&L pairs: economic relationship and credit-dominance tests, boundary cases included; forward-starting assessments run each reporting period. The designation memo (conceptual, not compliance advice) documents the archetype: a roaster/buyer patterned on how a branded coffee company describes its commodity hedging in its annual report.

8.3 EMIR 3: clearing thresholds

The problem and the regime

After 2009, G20 commitments pushed standardized derivatives to central clearing and reporting; EMIR implements that in the EU. EMIR 3 (Regulation (EU) 2024/2987, in force 24 December 2024 [5]) reshaped counterparty categorisation: financial counterparties (FC) and non-financial counterparties (NFC+, NFC-) are sorted by their derivative positions, and only counterparties exceeding *clearing thresholds* are forced to clear.

What changed and where this program sits

EMIR 3 moved the threshold calculation to *uncleared OTC positions only* (recognizing the benefit of clearing), computed at entity level for NFCs, with the *hedging exemption* retained: contracts that objectively reduce commercial risks of the counterparty or its group are excluded. ESMA's February 2026 final technical standards set the uncleared commodity and emission-allowance threshold at EUR 4 billion [6]. This program's annual notional is roughly EUR 3.6M ($\approx$ 0.009% of the threshold): the clearing obligation is nowhere near engaged, and the workbench's exchange-traded futures are cleared by construction. The mapping is directional context, not a classification claim.

8.4 MiFID II and PRIIPs: distribution and disclosure

A half-section because it only frames the note's boundaries. MiFID II governs derivative trading and sets position limits for commodity derivatives (with hedging exemptions); PRIIPs (Regulation (EU) No 1286/2014, [7]) requires a key information document before any structured product is offered to a retail investor. The participation note in Chapter 6 is priced two ways but offered to nobody; the KID obligation maps to zero here. Recorded so the boundary is explicit.

Part II
Validation and decision: what was checked and what it decided

9 Scope and Honest Framing

This project extends the Citi MQA Forage simulation onto real frozen market data. It is a learner workbench, not a production system: no Citi affiliation is claimed, no internal methodology is reproduced, no regulatory compliance is asserted, and nothing here is investment advice. The data (Yahoo Finance daily bars, SOFR from FRED) is checksummed and gated; exchange-data redistribution licensing forbids committing the price bars, so only manifests, gate reports and executed notebooks are committed and the bars regenerate locally through the gated freeze. Any future cloud artifact is gated snapshots only.

10 Program structure

The workbench was built in five methodology phases and two review phases. Part I explains each phase's methods chapter by chapter; the map below links phases to deliverables for navigation.

Phase	Method	Explained in
1. Data gates + MQA replication	SHA-256 manifest freeze, count/completeness gates, gold fallback, fail-closed; the Forage Task-3 methodology re-run on the frozen curve	Ch. 2 (2); outcomes in 12
2. Carry & curve calibration	Cost-of-carry identity, Schwartz–Smith fit, GARCH(1,1) vs EWMA	Ch. 3 (3), Ch. 4 (4)
3. Roaster hedging program	Futures + zero-cost collar, contract selection and roll, margin funding	Ch. 5 (5)
4. Structured note	Black-76 closed form vs antithetic Monte Carlo, bump-and-reprice greeks	Ch. 6 (6)
5. Risk gates + junior xVA	EE/EPE/PFE engine, bucketed unilateral CVA, netting and collateral	Ch. 7 (7)
6. Validation harness	SR 26-2-aligned inventory, no-look-ahead gate, rolling outcomes	12

Table 1: Phase map. The decision analysis and limitations close Part II.

11 Hand-Calculated Verification

Per the spec, an interviewer must be able to verify the numbers without reading code. Each check below is a test in the suite with the arithmetic worked out in its docstring.

12 SR 26-2-Aligned Validation

The posture is methodological fidelity, not compliance: the harness reproduces the *mechanics* SR 26-2 (2026-04-17, superseding SR 11-7) expects of model-risk discipline (inventory, conceptual soundness, outcomes analysis, benchmarking) over a frozen dataset, with every finding reproducible from the package.

Check	Hand calculation	Result
CVA, two buckets, constant EE \$1M, hazard 10%, LGD 60%, $r = 4\%$	$dS_1 = 1 - e^{-0.05} = 0.04877058$, $df_1 = e^{-0.02} = 0.98019867$; $dS_2 = e^{-0.05} - e^{-0.10} = 0.04639219$, $df_2 = e^{-0.04} = 0.96078944$; CVA $= 0.6 \times 10^6 \times (0.04780387 + 0.04457460)$	\$55,427.08 (exact match)
VaR coverage, two 5-point windows at 80%	W1 hist $[-5..-1]$: VaR $= \lvert q_{0.2}\rvert = 4.2$, next window 0 breaches; W2 hist $[-2..2]$: VaR $= \lvert q_{0.2}\rvert = 1.2$, next window ($-10$ first) 1 breach	VaR $[4.2, 1.2]$, breaches $[0, 1]$
EWMA forecast, returns $[1, 2, 3]\%$, $\lambda = 0.94$	weights $w = (1, 0.94, 0.8836)$; weighted variance $\frac{\sum w(x-\bar{x})^2}{\sum w - \sum w^2/\sum w} = 0.9993624$; annualised $\sqrt{252 \times 0.9993624}$	15.869%/yr
Collar net cost, synthetic zero-gap collar (f0 320, put 280, call 350), volume 100k lb	unhedged $= 1000 f_T$; futures locked at 320k; collar: 240k below the put ($f_T - K_p + f_0$), 320k between, 350k at $f_T = 380$	exact on all four scenarios
Filtered VaR (95%), 4-day price series 100/101/103/102, \$1000 per return-fraction	VaR at t uses only returns $\leq t-1$ and price($t-1$): EWMA var ($\lambda = 0.94$, weights 0.06/1) over $\{1\%, 1.9802\%\} = 0.054381/0.11321 = 0.48039\,\%^2$ $\rightarrow$ sd $0.6931\,\%$/day; VaR $= 1.6449 \times 0.006931 \times 103 \times 1000$	1,174.26 (exact match)
Collar margin cap, contracts = 2 (\$750/cent)	futures drawdowns 15k/90k; collar funding loss capped at $(320 - 280) \times 750 = 30\mathrm{k}$	$[15k, 30k]$ capped

Table 2: Hand-calculated verification table (all reproduced exactly by the test suite).

12.1 Model inventory

12.2 Conceptual soundness

Each model's assumptions are stated in its module docstring and reproduced in the inventory above. Two deliberate simplifications are flagged rather than hidden: the Schwartz–Smith fit is reduced-form on a single snapshot (volatility parameters would need the full Kalman MLE), and the collar's margin treatment assumes the long put's gain is monetisable day-by-day (real clearing charges full futures VM daily and settles options separately).

12.3 No-look-ahead gate

Calibration windows are gated fail-closed: any observation after the valuation as-of date raises a `LookaheadError`. The historical as-of reconstructions in the outcomes program pre-truncate each frozen series to the as-of window and then gate; an untruncated (contaminated) input is refused. The gate is itself tested against a poisoned window.

12.4 Outcomes analysis

The VaR coverage FLAG, and its remediation. The static historical 95% VaR undercovers by roughly 50% on this book: 38 breaches against 25.2 expected over four 126-day windows. Two causes were identified. First, the classic unconditional-historical-VaR failure mode on fat-tailed, volatility-clustered returns: the trailing window's empirical quantile understates the next window's tail. Second, a flaw in the backtest itself: holding one stale VaR constant across a 126-day test window conflates volatility-regime drift with model failure. The

Model	Inputs	Key assumptions	Benchmark	Outcomes check	Review
Data gates	frozen chain + manifests	count/completeness ceilings	SHA-256 checksums	fail-closed on injected gap	2026-09, green
Implied carry	frozen chain, SOFR	expiry ≈ 15th; storage $d = 0$	cost-of-carry identity	curve stability at past as-ofs	2026-09, green
Schwartz–Smith	curve snapshot	reduced form; κ weakly identified	NLS RMSE, $\kappa > 0$	RMSE vs tolerance	2026-09, green
GARCH(1,1) + EWMA	daily returns	constant mean	EWMA ($\lambda = 0.94$)	rolling forecast vs realized	2026-09, green
Black-76 + collar	F, K, vol, SOFR	European; discounted premiums	put-call parity; zero-cost gap	gap ≈ 0 by construction	2026-09, green
Participation note	F0, vol, SOFR, p	risk-neutral martingale futures	MC vs closed form	MC re-check	2026-09, green
IFRS 9 effectiveness	hedge P&L pairs, ratios	principles-based (no 80–125% band)	credit-dominance rejection	boundary tests	2026-09, green
Discrete-CVaR ratio	scenario P&L grid	discrete scenarios	MDPI (2025) hedge-ratio study	directional invariants	2026-09, green
GARCH-BEKK cross hedge	coffee/gold returns	full 2×2 BEKK(1,1), variance-targeted	literature hedge ratios	ratio stability	2026-09, green
Margin stress	frozen path, vol, program	monetisable-put funding cap	FSB 2024 framing; relief identities	peaks vs buffer	2026-09, green
Historical VaR/ES	realized P&L	empirical tail	ES ≥ VaR invariant	rolling breach count	2026-09, green
EE/EPE/PFE	F0, vol, SOFR, book	lognormal martingale	analytic forward MtM	invariants in suite	2026-09, green
Unilateral CVA	EE profile, hazard, SOFR	flat hazard/SOFR; no DVA	QuantLib re-derivation	CVA-vs-QuantLib re-quote	2026-09, green

Table 3: Model inventory (thirteen models, Phases 1–5; all reviewed 2026-09 with the suite green).

Check	Finding	Status
Working vol forecast	GARCH RMSE 4.01 vs EWMA 6.80 %/yr over 4 rolling windows (ratio $0.59 \leq 1.10$)	PASS
No-look-ahead gate	active on 8 frozen series (as-of 2026-09-04); rejects a poisoned as-of window	PASS
Curve stability	front implied yield 14.8–40.8 %/yr over 6 historical as-of dates; backwardated at 4 of 6 as-ofs (two historical contango episodes)	PASS
VaR coverage (95%), static historical	38 breaches vs 25.2 expected over 4 windows (ratio 1.51, band 0.5–1.5)	FLAG
VaR coverage (95%), filtered EWMA	37 breaches vs 33.7 expected over 673 days (ratio 1.10), remediating the static FLAG	PASS
Note MC vs closed form	$\lvert\text{diff}\rvert = 8.2 \times 10^{-5}$ per unit notional, s.e. 6.2×10^{-4} (≤ 2 s.e.)	PASS
CVA vs QuantLib	relative difference 1.9×10^{-16} (same discretization, tol 10^{-9})	PASS

Table 4: Outcome-analysis findings on the frozen data.

remediation is implemented in the package (ticket 13-05, `risk.filtered_var`): a filtered estimator pricing each day with the lagged EWMA variance ($\lambda = 0.94$) scaled by the lagged price. Re-run on the same book it covers at 5.5% of days versus the 5% expectation (37/673, ratio 1.10, inside the pre-declared band). The static historical estimator is retained in the findings as the baseline and its FLAG stands as the recorded evidence for why the filtered form is the production estimator of this workbench; the acceptance band was not moved.

13 Decision: Hedged vs Unhedged Outcome

Scenario engine: 50,000 lognormal martingale paths over one quarter (0.25y), the same distribution Black-76 prices. Buyer's economics: the roaster pays volume $\times f_T$ regardless; each hedge's terminal P&L reduces it. Note that the collar's strikes bound the *hedge P&L*, not the cost: between the strikes the cost is locked at the forward; below the put (280) and above the call (382.06) the net cost moves 1:1 with the market again: a crash below the put keeps helping the buyer, a spike above the call keeps hurting.

Strategy	Cost mean	Cost p95	Cost worst	Margin peak mean	Margin peak p95
Unhedged	3,895,696	5,288,885	10,296,484	—	—
Futures	**3,891,000**	3,891,000	3,891,000	43,655	103,797
Collar	3,893,639	4,595,138	9,602,737	30,647	44,250

Table 5: Annual purchase cost (1.2M lb) and margin-funding peaks (USD), 2.67 average contracts, put 280 / call 382.06, 0.25y horizon.

Reading the table. The full futures hedge is a strict price lock at \$3,891,000 (= 1.2M $\times$ 3.2425); its residual risk is purely margin funding: mean peak \$43.7k, 95th percentile \$103.8k, comfortably inside the \$800k reference-collateral assumption carried from the Starbucks 10-K framing. The collar costs almost the same on average (\$3.894M) but trades tail exposure for liquidity: its margin peak is capped at the put protection ($44{,}250 = (324.25 - 280) \times \frac{8}{3} \times 375$), roughly 58% lower at the 95th percentile, while the cost re-floats 1:1 beyond the strikes (p95 \$4.595M, worst \$9.6M in-simulation). The unhedged roaster owns the entire distribution: p95 \$5.29M, simulated worst \$10.3M.

Decision. For a roaster whose binding constraint is cash-flow certainty (the Starbucks 10-K template: \$37.9M of margin collateral posted as of FY2025), the choice is between the futures lock (price certainty, full margin exposure) and the collar (locked cost band, half the margin peak, tail re-exposure). The workbench's numbers support the collar for a margin-constrained buyer and the futures lock when budget certainty dominates, with the IFRS 9 cash-flow-hedge designation and effectiveness testing identical in structure for both.

14 Limitations

- **Basis risk is not modelled.** All hedges are evaluated on the front KC=F series; the roaster's physical purchases settle on a differential basis to the exchange. A basis-scenario overlay is the natural next step.
- **Historical volatility only.** The working vol is a GARCH(1,1) point estimate on frozen history; option premiums (collar legs, the note) inherit its parameter risk. No implied-vol surface is available for free coffee options.
- **VaR remediation is EWMA-level.** The static historical VaR's under-coverage is remediated by the filtered (EWMA $\lambda = 0.94$) estimator (Section 12); a GARCH-conditional-vol scaling would be the next refinement but is not implemented. The static estimator is kept only as the flagged baseline.
- **Unofficial source.** The Citi MQA Forage simulation is a learning artifact, not a source of Citi methodology; all toy numbers from it are labelled context only.
- **Junior xVA only.** Unilateral CVA with flat hazard and flat SOFR; no DVA/FVA/MVA, no wrong-way-risk modelling beyond a qualitative discussion (a buyer short a call on coffee is long the reference asset; WWR here is benign by inspection).

- **Margin conventions are assumptions.** $8k initial margin per contract and the monetisable-put funding cap are stated learner assumptions, not exchange feeds.

References

[1] Black, F. (1976). The pricing of commodity contracts. *Journal of Financial Economics* 3(1–2), 167–179.

[2] Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics* 31(3), 307–327.

[3] Board of Governors of the Federal Reserve System, OCC, and FDIC (2026). *Supervisory Guidance on Model Risk Management*, SR 26-2 attachment, April 17, 2026.

[4] International Accounting Standards Board (2014, as amended). *IFRS 9 Financial Instruments*, Section 6, Hedge Accounting. IFRS Foundation.

[5] European Union (2024). Regulation (EU) 2024/2987 (EMIR 3), amending Regulation (EU) No 648/2012, in force 24 December 2024.

[6] European Securities and Markets Authority (2026). *Final Report on draft RTS amending Commission Delegated Regulation (EU) No 149/2013 to further detail the new EMIR clearing thresholds regime*, ESMA74-1049116226-944, February 2026.

[7] European Union (2014). Regulation (EU) No 1286/2014 on key information documents (PRIIPs).

[8] RiskMetrics Group (1996). *RiskMetrics Technical Document*, 4th edition. J.P. Morgan/Reuters, New York.

[9] Kupiec, P. H. (1995). Techniques for verifying the accuracy of risk measurement models. *Journal of Derivatives* 3(2), 73–84.

[10] Schwartz, E. S., and Smith, J. E. (2000). Short-term variations and long-term dynamics in commodity prices. *Management Science* 46(7), 893–911.

[11] Hull, J. C. *Options, Futures, and Other Derivatives.* Pearson (multiple editions).

[12] QuantLib contributors. *QuantLib: a free/open-source library for quantitative finance.* `https://www.quantlib.org`.

[13] Federal Reserve Bank of St. Louis (2026). Secured Overnight Financing Rate (SOFR), series SOFR. FRED. `https://fred.stlouisfed.org/series/SOFR`.